

Recitation: Partial Fractions

Problem 1. Find the partial fractions decomposition for the following:

(a) $\frac{x + 13}{x^2 - 4x - 5}$

(b) $\frac{3x + 13}{2x^2 + 4x - 6}$

(c) $\frac{5x^2 - 3x + 4}{(x - 1)(x^2 + 1)}$

Problem 2. Find the partial fraction decomposition of the following:

$$\frac{3x^4 - x^3 + 7x^2 - 4x + 2}{x^3 + 2x}$$

Problem 3. Complete the following:

(a) Find the partial fractions decomposition of $\frac{x^2 - x + 2}{x^3}$ ‘immediately.’

(b) Find a rational function with the partial fraction decomposition $2 + \frac{-3}{x+1} + \frac{1}{x-2}$.